

What's an anthropologist doing in Meta's Infrastructure?

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Researcher, Meta



“Ptimothy?”

@will_santino_illustration





Arthropod $\neq$ Anthropology





Archaeology is a form of anthropology



Meta AI



Forensic anthropology is also a form of anthropology



An isometric illustration of a networked office environment. In the center, a grid of nodes (circles) is connected by lines, representing a network. Several stylized human figures are positioned around and within this network, some appearing to interact with it. The background is filled with various office equipment and machines, including large computer cabinets, a printer, a scanner, and a monitor displaying text. The overall style is a detailed, hand-drawn isometric illustration. The text "Improving tools and policies by giving attention to culture" is overlaid in the center in a large, white, sans-serif font. The text "Meta AI" is visible in the bottom right corner of the illustration.

**Improving tools and policies by
giving attention to culture**

An open, weathered metal toolbox sits on a wooden workbench in a workshop. The toolbox is filled with various tools, including a pair of pliers and a screwdriver. The background shows shelves with more tools and a blue storage bin. A black rectangular box with the white text 'UXR' is overlaid on the toolbox.

UXR

The background of the image is a workshop. In the foreground, a metal toolbox with a handle and latch sits on a wooden workbench. The toolbox is open, and its interior is visible. In the background, there are shelves with various tools and materials, creating a sense of a well-used workspace. The lighting is warm and focused on the toolbox.

Mixed Methods

Policy Analysis

UXR

Intervention Design

Future Forecasting

Experimental Design



Meta AI



Meta AI



1. Recognizing reliability work



**1. Recognizing
reliability
work**



**2. Making
configuration
changes safer**



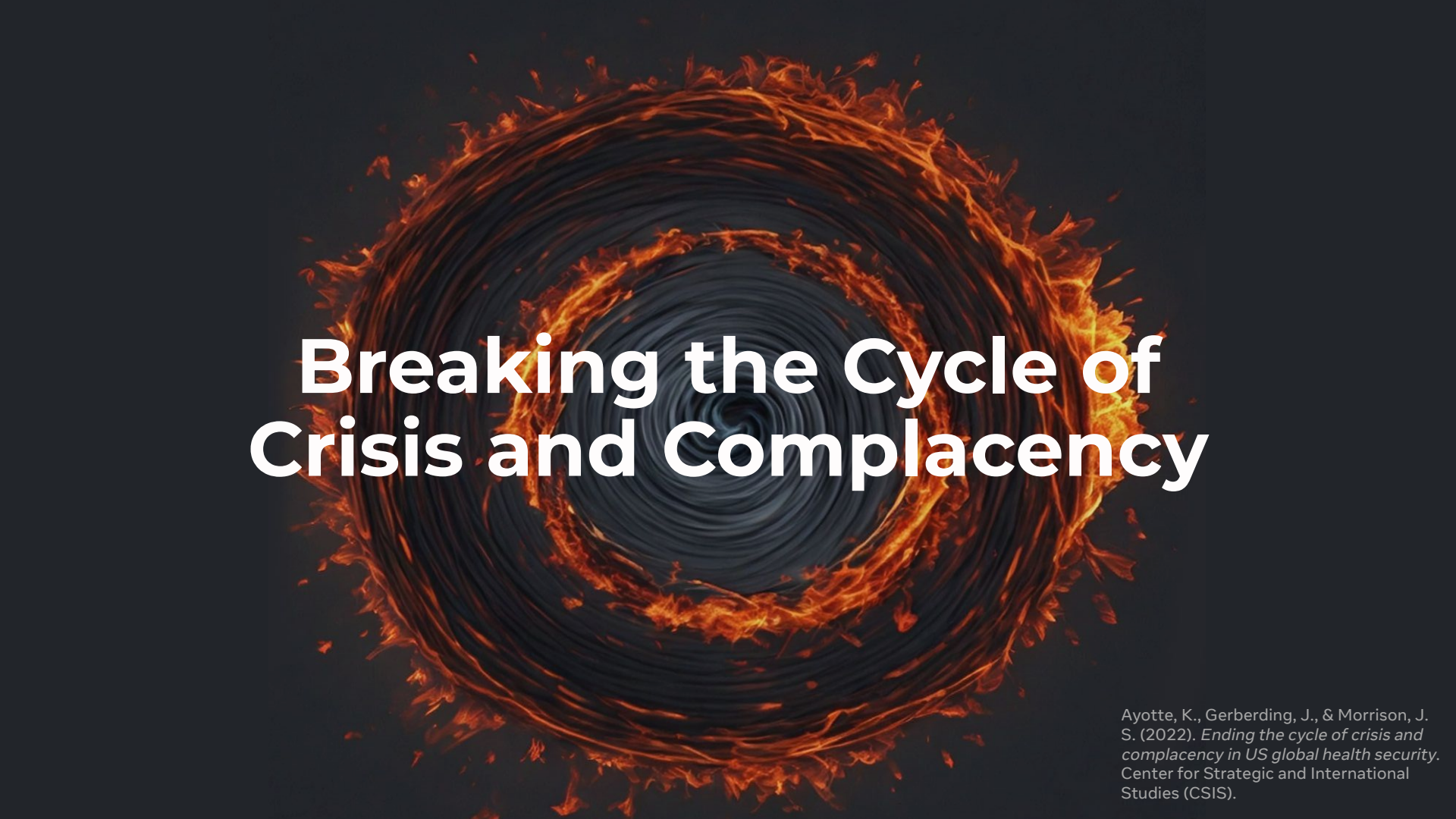
**1. Recognizing
reliability
work**



**2. Making
configuration
changes safer**



**3. Testing AI
for risk
reduction**



Breaking the Cycle of Crisis and Complacency

Ayotte, K., Gerberding, J., & Morrison, J. S. (2022). *Ending the cycle of crisis and complacency in US global health security*. Center for Strategic and International Studies (CSIS).

Stone agers in the fast lane



A close-up photograph of a person's face, with their hands covering their eyes. The image is dimly lit with a blue and purple color cast. The text "People are not great at assessing risk" is overlaid in white, bold, sans-serif font.

**People are not great
at assessing risk**

**We're bad at assessing
slow-burn threats**



Meta AI

$KPI = \emptyset?$



Preparing for risk in a fast-paced world

**MOVE
FAST AND
BREAK
THINGS**

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**MOVE
FAST AND
BREAK
THINGS**

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MESSANGER



NEWS FEED



WORKPLACE



SAFETY CHECK

VIDEOS

**MOVE
FAST WITH
STABLE
INFRA**

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The background of the slide is a blurred photograph of two individuals. In the foreground, the back of a person's head with dark hair tied in a bun is visible. In the background, another person is partially visible, looking towards the camera. The overall scene suggests an interview or research setting.

Foundational research through semi-structured interviews



Extroverts@Meta: 97 members



Extroverts@Meta: 97 members

Introverts@Meta:



Extroverts@Meta: 97 members

Introverts@Meta: 4,193 members

The background of the slide is a blurred photograph of two individuals. In the foreground, the back of a person's head with dark hair tied in a bun is visible. In the background, another person is partially visible, looking towards the camera. The overall scene suggests an interview or research setting.

Foundational research through semi-structured interviews

Systematizing Qualitative Data Analysis

The image displays a qualitative data analysis software interface. On the left, a 'Code System' panel shows a hierarchical list of codes with their respective counts. An orange box highlights this panel, and an orange arrow points from it to a larger view of the same panel on the right. The main interface on the right features a 'Document Browser' showing a list of documents and a 'Retrieved Segments' panel at the bottom.

Code System Hierarchy:

Code System	Count
Code System	585
Oncall Health	6
Dependencies of Reliability Work	22
Defining Reliability	35
General Reliability Sentiment	32
Disincentives to do Reliability Work	25
Incentives to do Reliability Work	15
Recognizing Reliability Work	20
Awareness of Reliability Efforts	2
Team-based Reliability Efforts	53
Difference in valuing reliability	36

Document Browser: ID 12 SWE Core Workflows (168 Paragraphs)

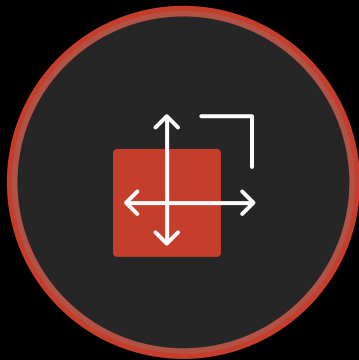
Retrieved Segments:

Activate documents and codes to retrieve coded segments ...

Text Excerpt:

Yeah. I mean, we've got a weekly reliability meeting. We've just actually, as I just mentioned, we joined this bigger org. Well, we were in a bigger org too, but we moved to another org. And this org has a kind of weekly SEV review for just our org. And kind of a little bit is redundant to our preexisting reliability meeting we have weekly. But we're still keeping both because it is, like I said, it's kind of a major thing for us. And it's always been in a sense like at least 30%, I guess, of our efforts. Maybe more has been always trying to improve reliability. I mean, maybe that speaks to maybe we did start off very reliable. Actually, I think we've been up pretty good. But the thing is, it's always like, as you can imagine, there's always your initial kind of plans. What's that Mike Tyson quote? "Everybody's got a plan until you're

Key Themes



Reliability is necessary
the greater we scale



We want reliability to
facilitate innovation,
not block it



Our current system
depends on reliability
champions



Demonstrating the impact
of reliability work and
rewarding that impact

"I love reliability and I'm frustrated by doing it. It's not as recognized, it's difficult, it requires a lot of work, and the tooling is really bad."

SOFTWARE ENGINEER

A close-up, slightly blurred photograph of a person's hands writing on a white sheet of paper with a dark pen. The person is wearing a light-colored, long-sleeved button-down shirt. The background is dark and out of focus. Overlaid on the center of the image is white text.

**Surveys to get broad insights and
compare sentiment over time**

OVERALL SENTIMENT

78%

SAY THEIR TEAM
VALUES
RELIABILITY

70%

AGREE TEAMS
SHOULD DO MORE
RELIABILITY WORK

54%

FIND IT HARD TO
IDENTIFY
RELIABILITY GAPS

TOP RELIABILITY BARRIERS

64%

COMPETING
PRIORITIES

53%

SHOWING
RELIABILITY
IMPACT

40%

COMPLEXITY OF
THE SYSTEM /
DEPENDENCIES

WAYS TO IMPROVE RELIABILITY

1 RECOGNITION & INCENTIVES
2 TOOLING & RESOURCES

3 CULTURE CHANGES
4 BETTER PRIORITIZATION

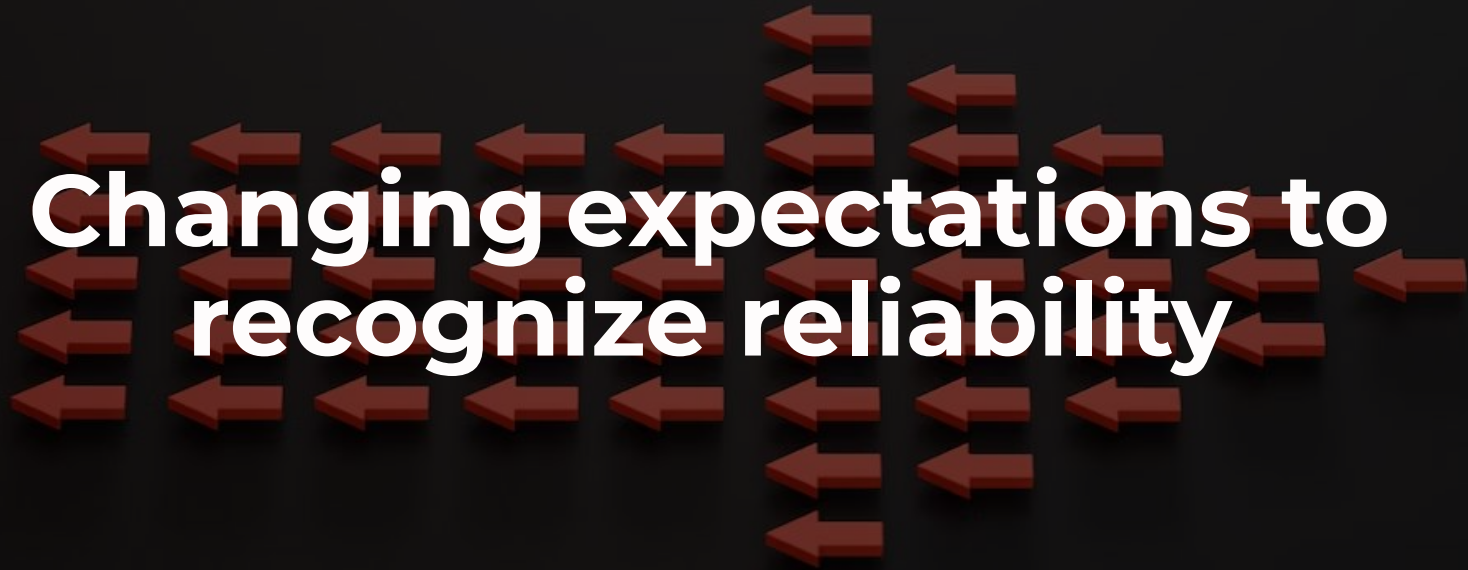
FROM OPEN-ENDED COMMENTS

RELIABILITY

AT META

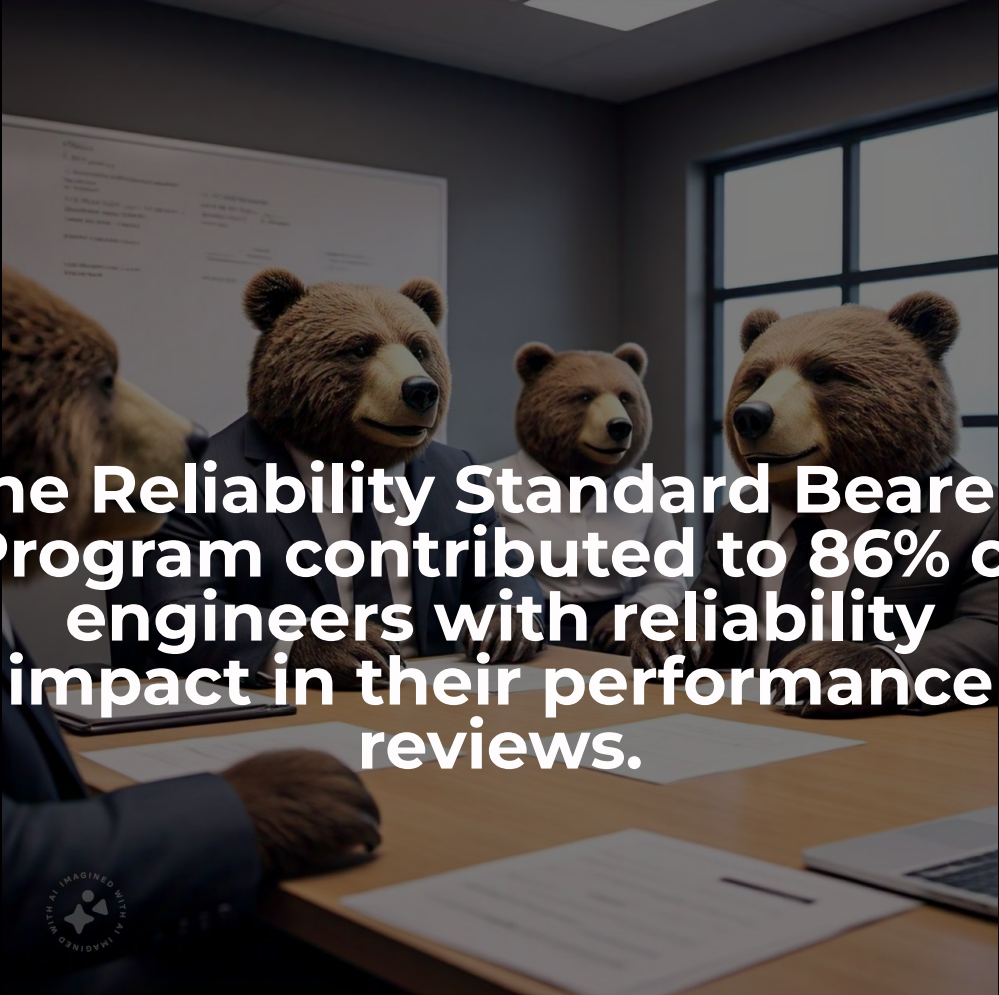


1. Recognizing reliability work



**Changing expectations to
recognize reliability**



A surreal image of four anthropomorphic brown bears wearing business suits and ties, sitting around a wooden conference table in a modern office. They appear to be in a meeting, with papers and a laptop on the table. In the background, there is a whiteboard with some faint text and a large window with a grid pattern. The scene is dimly lit, with light coming from the window.

**The Reliability Standard Bearers
Program contributed to 86% of
engineers with reliability
impact in their performance
reviews.**





55%

**FEEL SATISFIED
WITH HOW
RELIABILITY
WORK IS
RECOGNIZED IN
PERFORMANCE
REVIEWS**

55%

FEEL SATISFIED
WITH HOW
RELIABILITY
WORK IS
RECOGNIZED IN
PERF@





2. Making configuration changes safer

MAR. 5, 2024

**Facebook, Instagram, and Threads
were
all down**

APR. 3, 2024

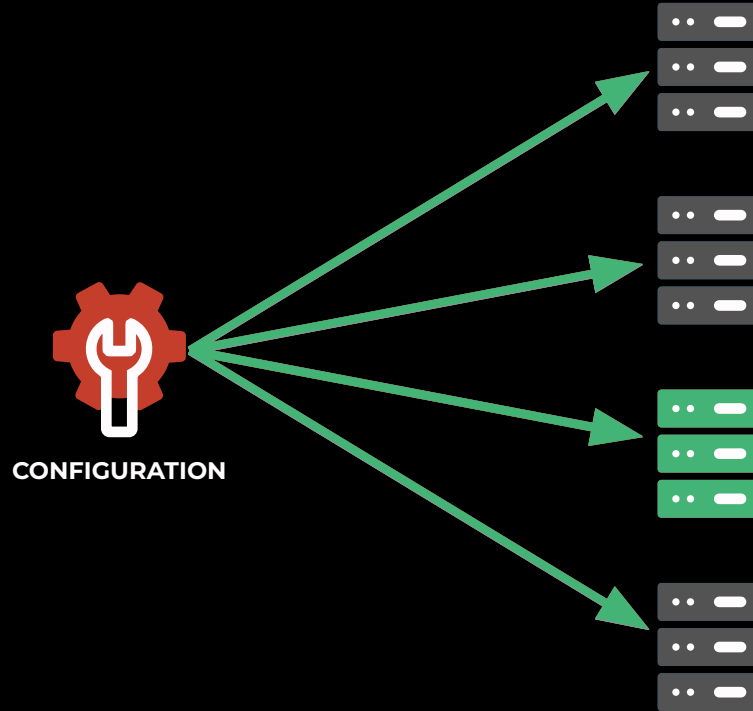
**WhatsApp, Instagram, and Facebook
are back up and running after outages**

A large orange bracket graphic that frames the text on the left side of the image.

52%

**FEEL CONFIDENT
THAT IN THE
PROCESSES TO
PREVENT BAD
CODE AND
CONFIG
CHANGES**

Improving configuration tooling





Change Safety 101

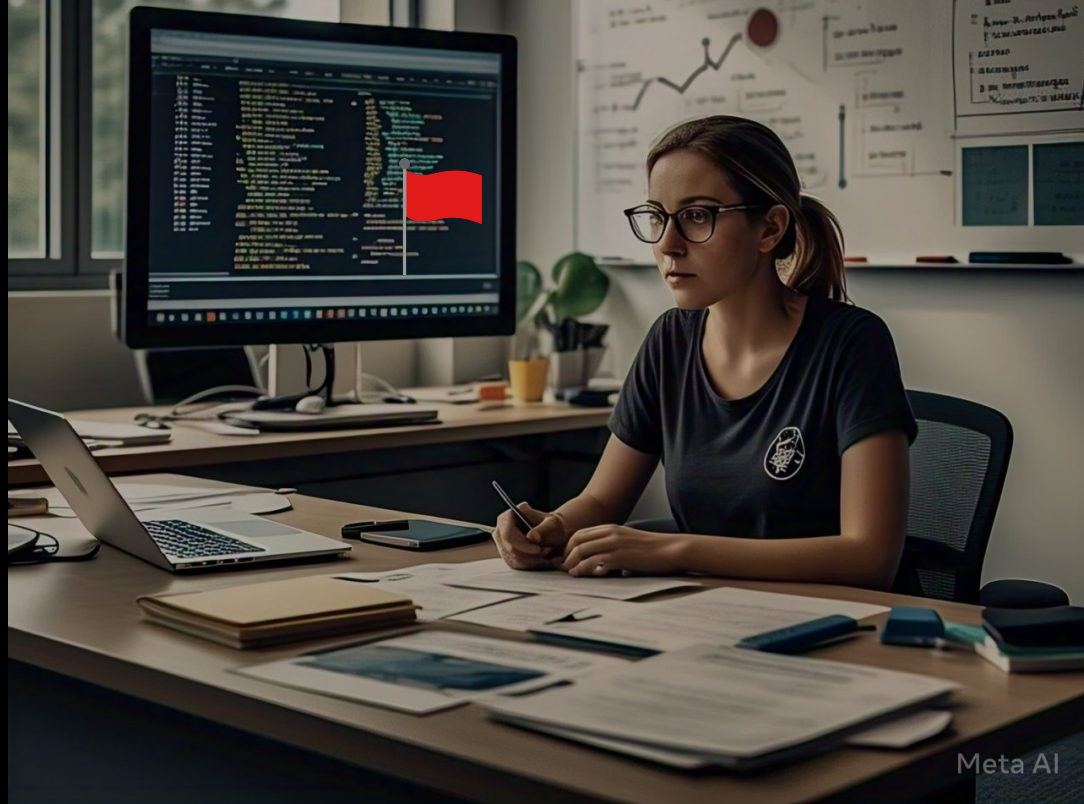
Welcome to Change Safety 101

You will learn the basics of change safety, code deployment, config canaries, health checks, and gating code. If you have any questions, don't hesitate to reach out at

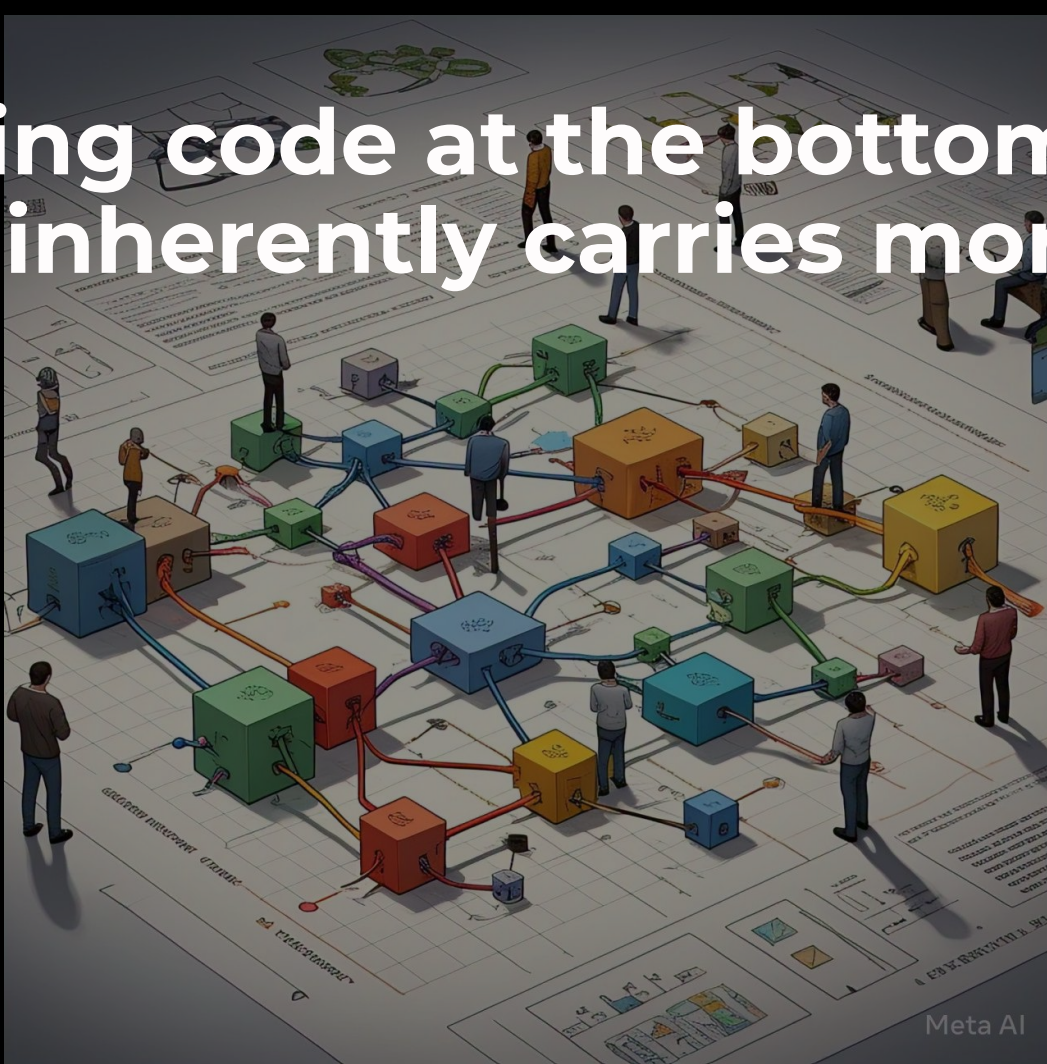


3. Testing AI for risk reduction

Using AI to flag risky code changes



Authoring code at the bottom of the stack inherently carries more risk



**Don't leave AI out in the wild
unattended**





**Regularly run
disaster
recovery drills
to simulate
the loss of
our data
centers**



**Formalized
Reliability
Measurement
Program to
get quality
top-line SLOs**



**Created a
Maturity
Model to
guide
Reliability
Programs and
prioritize
action**

A large orange bracket graphic that frames the text on the left side of the image.

91%

**SAY THEIR
TEAMS
VALUE
RELIABILITY**

A look at Meta's planetary-scale computing infrastructure, including some key lessons from its development, as the company pursues its vision of "all global datacenters as a computer."

BY CHUNQIANG TANG

Meta's Hyperscale Infrastructure: Overview and Insights

HYPERSCALERS, SUCH AS Alibaba, Amazon, ByteDance, Google, Meta, Microsoft, and Tencent, have developed

Kubernetes and PyTorch originated in Google and Facebook, respectively, but have been adopted by organizations of all sizes. In addition to these specific technologies, the principles and lessons from hyperscale infrastructure may assist practitioners in building better systems in general.

This article provides a high-level overview of Meta's hyperscale infrastructure, focusing on key insights from its development, particularly in systems software. Where relevant, we highlight differences from public clouds, as varying constraints have led to distinct optimizations. Though much of the knowledge presented here has been shared and practiced within the industry and research community, including insights from our past publications, the article's primary contribution is to provide a holistic perspective that helps readers build a comprehensive mental model of hyperscale infrastructure end to end.

Engineering Culture

Before delving into the details of Meta's infrastructure, we first highlight several aspects of the company's engineering culture, because an organization's culture heavily influences its technology.

Move fast. Since its inception, Face-

» key insights

- Meta's engineering culture emphasizes moving fast, technology openness, research in production, and shared

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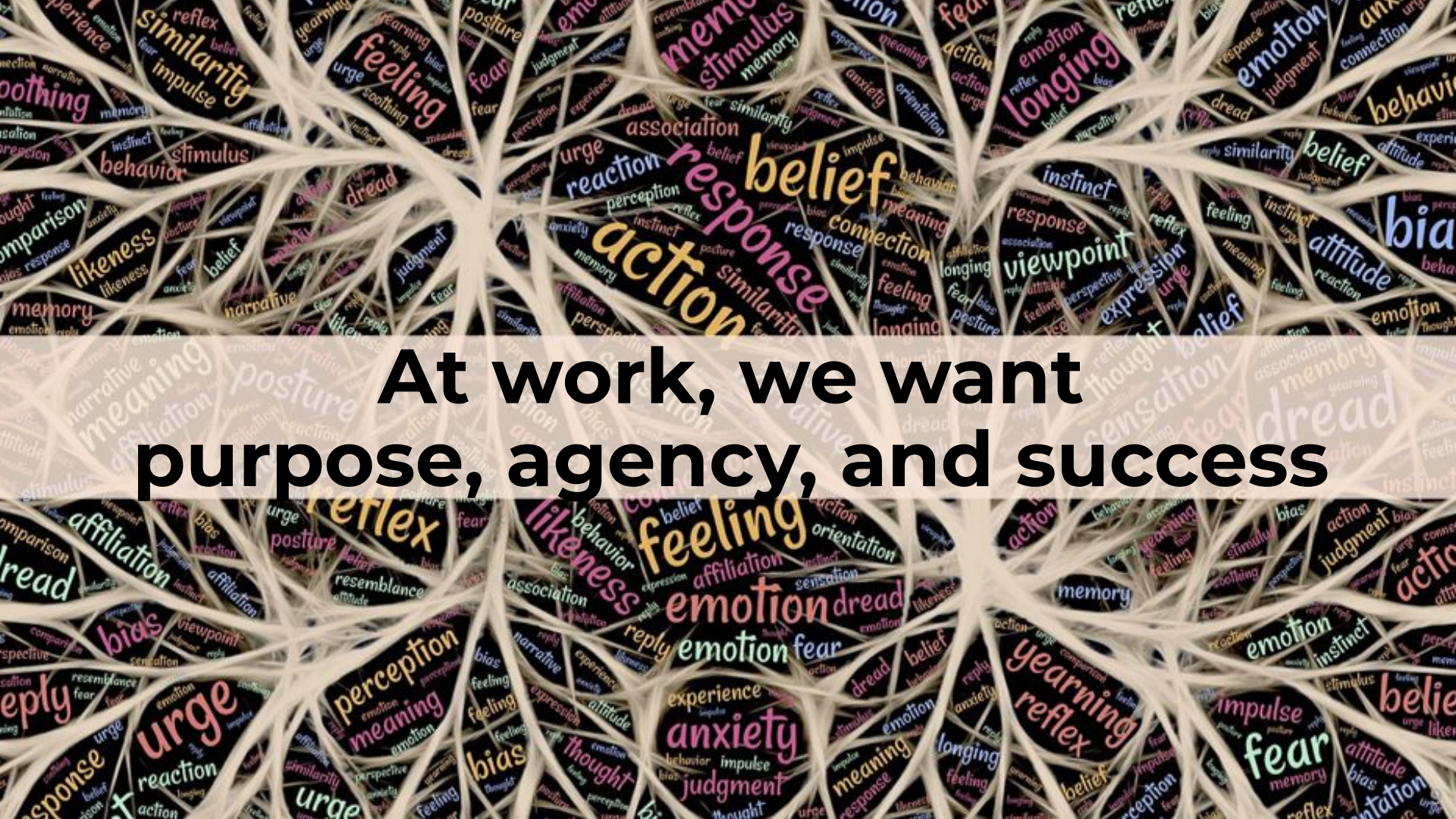
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**At work, we want
purpose, agency, and success**



**As we innovate, use culture to
support and stabilize**

Recap

Interviews

Listen to what matters on the ground

Surveys

Gather high-level, lightweight insights over time.

Actions

Use cultural data in conjunction with risk analyses to guide actions and priorities.

Balance

Be mindful of heavy, top-down initiatives and test the efficacy and acceptance of new interventions

Thank you

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